INVENTORY RECONCILIATION AGENT · MIPS · ODI · WMS · ORACLE

Simulation Dummy Data

What the synthetic dataset contains, why each row exists, and how it drives the live simulation

Companion workbook	Simulation_Dummy_Data.xlsx (same folder)
Simulated sales day	Saturday 07 November 2026
Plant	UBD · Lehigh Valley PA Bakery (invented values)
Source code	recon_agent/events/seed.py, scenarios.py, contract.py, derive.py
Document date	03 November 2026

All data in this pack is synthetic.

SKUs are numbered 900101-900112 and every item and customer name starts with "SIM".

The real client extracts in data/ are not part of this pack.

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1. Purpose and scope

The Inventory Reconciliation Agent compares four source systems - **MIPS** (planning), **ODI / Oracle production**, **WMS** (warehouse) and **Oracle Fusion** (inventory postings) - and flags every mismatch with a quantified correction. To demonstrate this in real time without touching plant systems, the project ships a simulation: a source simulator publishes row-level changes as events, and the worklist reacts to them.

This document describes all of the dummy data behind that simulation:

- **The starting day (seed):** one invented Saturday sales day, built so that each SKU shows one known reconciliation problem. It publishes 97 insert events across 8 tables.
- **The scenario library (S1-S8):** eight scripted stories, one per requirement scenario, that change the starting day live and make new flags appear.
- **The expected results:** the flags the reconciliation engine raises on that data, with units and dollar values.

> **Out of scope.** The five Excel files in `data/` (MIPS Order Data, ODI Batch and Finished Good Data, WMS Fill Rate and Order Summary reports for 12 Sep 2026) are real client extracts used by the batch mode. They are not dummy data and are not reproduced here.

1.1 Companion workbook

`Simulation_Dummy_Data.xlsx` holds every row described here: one sheet per source table with all contract columns, plus the SKU master, customers, the 14 starting-day flags, the scenario library, every scenario change, and a data dictionary. Section 11 is a sheet-by-sheet guide. Section 6 describes every rule.

2. How the dummy data is produced

The data is not stored as a file. Code generates it on demand, so it is always consistent with the event contract and the rules:

File	Role
recon_agent/events/seed.py	Builds the starting day: SKUs, customers, and every row of the 8 tables. Declares the 14 flags it must raise (EXPECTED).
recon_agent/events/scenarios.py	The eight scenarios as scripts of inserts and updates, each declaring the flags it must raise.
recon_agent/events/contract.py	The event contract v2: the 8 entities, their columns, types, keys and required columns.
recon_agent/events/derive.py	Columns a source system computes itself (e.g. Short Qty, stack and tray counts), recomputed on every change.
output/sim/events.db	The SQLite event log the seed and the scenarios publish into (the live simulation state).

2.1 Properties

- **Deterministic.** Every run of the seed produces identical rows (only wall-clock delivery times differ). The workbook was produced by seeding a throwaway log, so the live `output/sim/events.db` was not changed.
- **Clearly synthetic.** SKUs 900101-900112, BUSS codes 009101-009112, formulas SU9001xx, order numbers 99000xx, and every description or customer name starts with "SIM". A unit test enforces this.
- **Shaped like the client extracts.** Same column names and order, and the same quirks (Section 10), so the same ingest code reads events and Excel files.
- **Self-checking.** The test suite asserts that the seed raises exactly its 14 designed flags and that each scenario raises exactly the flags it declares. The scenario library is the rules' regression suite.

- **No faults in the seed.** The seed bypasses injected integration faults; scenarios do not, so a delay or drop fault applies to their changes.

2.2 Regenerating it

```
python -m recon_agent events seed --reset # publish the starting day, start the clock
python -m recon_agent events scenario list # the eight scenarios
python -m recon_agent events scenario run S6 --reset
python -m recon_agent events show wms.item_order # current rows of one table
```

The source simulator (`python -m recon_agent simulate`, port 8100) shows the same tables, lets an operator edit any row, runs scenarios, and has a *Reset to starting day* button.

3. The simulated plant day

The data describes the Saturday 07 November 2026 sales day at the UBD (Lehigh Valley) bakery. The plant clock starts at **20:00 on the Friday evening**: production is finished, WMS has allocated orders and is loading trailers. The timeline the rows encode:

Plant time	What happened	Where it shows
Tue 03 Nov 15:32	OCloud supply plan written to MIPS	MIPS lines, Source System = OCloud Region2
Wed 04 Nov 06:00-20:00	Day-shift production runs (Lines 1 and 2)	ODI finished goods and batches
Thu 05 Nov 00:00-12:00	Night and day runs, incl. the 900102 recovery run and the moved 900105 run	ODI finished goods and batches
Thu 05 Nov 20:31	BUSS sales orders last synced into MIPS	MIPS lines, Source System = BUSS
30 min after each run	WMS receives the run into the dock (ticket + receipt transaction)	WMS tickets, WMS transactions
2 min after each receipt	Oracle Fusion posts the receipt	Oracle inventory postings
Fri 06 Nov 14:00	WMS item order lines taken and allocated	WMS item orders
Fri 06 Nov 16:00-19:00	Order headers: North loaded and shipped, Route 12 shipped without a load scan	WMS orders
Fri 06 Nov 20:00	Simulation clock starts	Simulator top bar / worklist 'as of'
Sat 07 Nov	Sales day (Schedule Date / SalesDate on every row)	All tables

Scenario steps are stamped with the plant time at which they run (20:00 onward after a fresh seed). S4 and S8 advance the clock by 90 and 11 minutes respectively.

4. Master data

4.1 SKUs

Twelve invented bakery items. Ten each carry one designed problem; two are clean so that the scenarios can break them live and the new flag is easy to see.

SKU	BUSS	Description	Price	Units / tray	Designed problem	Flags
900101	009101	SIM 100% WW BREAD 24Z	$4.69	10	Inventory applied to the first of three order lines only	R01
900102	009102	SIM POTATO BREAD 22Z	$4.69	10	Recovery run over-built by 1,200	R05, R14

SKU	BUSS	Description	Price	Units / tray	Designed problem	Flags
900103	009103	SIM WHITE ROUND TOP 20Z	$3.99	10	Inventory netted with no beginning inventory recorded	R02
900104	009104	SIM WHITE ROUND TOP 2P 40Z	$5.49	6	Zero-quantity (blind) line in a live run	R03
900105	009105	SIM SQUARE WHEAT 24Z	$4.29	10	Run moved to Line 2; ghost copy left on Line 1	R04
900106	009106	SIM HOT DOG BUNS 8P 12Z	$3.59	12	WMS filled 700 units from untracked (freezer) stock	R07
900107	009107	SIM HAMBURGER BUNS 8P 12Z	$3.59	12	WMS order raised after MIPS synced; allocation short	R09, R10
900108	009108	SIM HONEY WHEAT 20Z	$4.49	10	400 surplus not carried to the next sales day	R06
900109	009109	SIM ITALIAN BREAD 20Z	$3.99	10	Units per container: 10 in MIPS, 12 in WMS	R11
900110	009110	SIM MULTIGRAIN 24Z	$4.79	10	Negative net order: inventory exceeds demand	R08
900111	009111	SIM BUTTER TOP 20Z	$3.99	10	Clean control (scenarios S1, S3, S5, S7)	-
900112	009112	SIM CINNAMON RAISIN 16Z	$5.29	12	Clean control (scenarios S2, S4, S6)	-

4.2 Customers

Customer	Order Nbr	Order type	Route	State at clock start
SIM DC NORTH	9900001	TRANSFER_ORDER	-	Loaded and shipped
SIM DC SOUTH	9900002	TRANSFER_ORDER	-	Packed; waiting for trailer (S8 ships it)
SIM DSD ROUTE 12	9900003	DSD_SALES_ORDER	3770012	Shipped with no trailer load scan (R12)
SIM DSD ROUTE 27	9900004	DSD_SALES_ORDER	3770027	Packed; waiting for trailer

Plant code UBD and the plant names follow the client's convention; all quantities, times and prices are invented.

5. The eight source tables

Each table is one entity in the event contract. The first five mirror the client's daily extracts column for column; the last three carry real-time detail no extract has, needed by rules R18 and R19.

Entity	System	Mirrors	Key	Rows
mips.order_line	MIPS	MIPS Order Data	Order Line ID (synthetic)	17
odi.finished_good	ODI	ODI Finished Good Data	FGEntryID (synthetic)	14
odi.batch	ODI	ODI Batch Data	BatchEntryID (synthetic)	6
wms.order	WMS	WMS Fill Rate Report	Order Nbr	4
wms.item_order	WMS	WMS Order Summary by Item	Item Line ID (synthetic)	23
wms.ticket	WMS	(real-time only)	Ticket Nbr	11
wms.transaction	WMS	(real-time only)	Txn Nbr	11
oracle.inventory_txn	ORACLE	(real-time only)	TransactionId	11

A *synthetic* key is a row ID the client extract does not have; the contract adds it so a change event can name the row it updates (an assumption to confirm with Bimbo IT).

5.1 mips.order_line (MIPS) · 17 rows

One MIPS schedule line: either a BUSS customer sales order or an OCloud supply-plan line. SKU demand for the day is the sum of its lines.

- 7 lines come from BUSS and 10 from OCloud Region2.
- ML-0001 is a BUSS order moved from the prior sales day: Order Moved = TRUE and its offset/BUSS sales dates are Fri 06 Nov while Schedule Date stays Sat 07 Nov. It is listed first for 900101.
- BUSS lines carry the FG Oracle Ref with a trailing space ("900101 "), as in the real extract (R15).
- ML-0007 has Order Quantity 0 (900104's blind line).
- ML-0013 (900109) says 10 units per container; WMS says 12 (R11).

5.2 odi.finished_good (ODI) · 14 rows

One ODI / Oracle production result for one SKU in one run: what MIPS asked for (NetUnitsOrdered), what was wrapped, and MIPS's view of inventory and order quantity.

- Runs that share a formula share an OracleTypeCode (SU900130 = 900103 + 900104; SU900160 = 900106 + 900107).
- FG-0003 is the 900102 recovery run (Line 2, Shift 3) ordering 2,000 after a 800-unit shortfall.
- FG-0006 is the zero copy left on Line 1 with no shift when run SU900150 moved to Line 2 as SU900150A (FG-0007).
- FG-0008 wrapped 3,300 of 4,000; FG-0010 wrapped 2,900 of 2,500.
- FG-0012 has NetUnitsOrdered -300 and MIPSBeginningInventoryQty 1,500.
- Derived: ExpectedEndingInventoryQty = wrapped - ordered; TotalPieces; TotalActualDamagedUnits.

5.3 odi.batch (ODI) · 6 rows

One mixing record per formula run: scheduled versus actual batches and the piece variance.

- Six batches; all mix as scheduled except BT-0003 (SU900121, the 900102 recovery run): 4 mixed against 3 scheduled (R14).
- Theo Batch Pieces Qty = pieces per batch x batches mixed. Derived: Final Piece Variance %.

5.4 wms.order (WMS) · 4 rows

One WMS customer order header with its fill and shipping progress.

- Order Qty and Allocated Qty are the sums of that customer's item lines at seed time.
- 9900003 (SIM DSD ROUTE 12) shows Shipped 7,900 with Loaded 0: no trailer load scan (R12).
- Derived: Short Qty, % Allocated, % Loaded, % Shipped, Progress (GREEN/YELLOW). Header totals are deliberately not re-derived from item lines, because a disagreement between them is rule R13.

5.5 wms.item_order (WMS) · 23 rows

WMS order quantity and allocation for one item and one customer.

- WL-0015 (900107, Route 27): Original Ord Qty 800, Order Qty 1,000, Allocated 800 (R09, R10).
- WL-0018 (900109) uses 12 units per container (R11).
- 900104 has no WMS lines at all: it has no demand.
- Derived: stack and tray counts (21 trays per stack, an assumption), units remaining, Completion% (GREEN when fully allocated, YELLOW otherwise).

5.6 wms.ticket (WMS) · 11 rows

A WMS receipt ticket: finished goods received from a plant run into the dock.

- One ticket per run that wrapped product (11; the three zero-output runs get none).
- Lot number = L + receipt date (yymmdd) + last 4 characters of the run code.
- TK-0010 (900111) is the ticket scenario S5 returns 120 units against. Derived: Net Received Qty.

5.7 wms.transaction (WMS) · 11 rows

A WMS inventory transaction that Oracle must post: receipt, ship, return to plant, freezer move or adjustment.

- The seed has one RECEIPT per ticket (WT-0001 to WT-0011), by user SIM RECEIVING.
- Scenarios add SHIP (S8), RETURN_TO_PLANT (S5) and FREEZER_OUT (S7) transactions.

5.8 oracle.inventory_txn (ORACLE) · 11 rows

An Oracle Fusion inventory transaction posted from a WMS transaction. SourceReference holds the WMS Txn Nbr.

- Every seed receipt is POSTED two minutes later (OT-0001 to OT-0011), so nothing is late at clock start.
- S8 adds one ERROR posting and leaves one WMS shipment with no posting: both become R18 flags after 10 plant minutes.

5.9 Column reference

The tables below describe the columns that matter in each source table: the row key, the columns that link it to other tables, the quantities the rules compare, and the derived columns the source system computes itself. **Used by** names the rules that read the column; a dash means the agent carries the column but no rule uses it. The workbook's Data Dictionary sheet lists every column.

mips.order_line (MIPS)

One schedule line: a BUSS customer sales order or an OCloud supply-plan line. A SKU's demand is the sum of all its lines.

Column	Role	What it holds	Used by
Order Line ID	Key (synthetic)	ML-0001 to ML-0017. Names the line in a change event.	-
FG Oracle Ref	Link: SKU	Oracle FG number, e.g. 900101. BUSS lines add a trailing space ("900101 ").	All SKU rules, R15
FG BUSS Code	Link: legacy code	009101 to 009112. Same value as ODI LegacyFGNumber. Not used to match.	-
FG Description	Attribute	The SIM item name.	Display
Source System	Attribute	BUSS = customer sales order. OCloud Region2 = supply-plan line.	R01, R16
Order Quantity	Quantity	Units demanded on this line.	R01, R02, R05-R09
Schedule Date	Date	The sales day (Sat 07 Nov) on every line.	KPIs
Offset Sales Date, Order Moved	Date / flag	A moved line keeps its Schedule Date, but its Offset Sales Date is the prior day and Order Moved = TRUE.	R07, R09
Production Date	Date	Empty on BUSS lines; sales day - 1 on OCloud lines.	-
Container Code, Units per Container	Attribute	Units per tray. Lines with container code 01 set the MIPS value.	R11
Retail Price	Attribute	Dollar value of one unit.	Flag $ values
Last Modify Date Time	Derived	Stamped with the plant time of every change to the line.	R09 evidence

odi.finished_good (ODI)

One production result for one SKU in one run: what MIPS asked for, what was wrapped, and MIPS's view of inventory and order quantity.

Column	Role	What it holds	Used by
FGEntryID	Key (synthetic)	FG-0001 to FG-0014. One row per SKU per run.	-
OracleFGNumber	Link: SKU	Same value as MIPS FG Oracle Ref and WMS Item Nbr.	All SKU rules
LegacyFGNumber	Link: legacy code	Same value as MIPS FG BUSS Code. Not used to match.	-
OracleTypeCode	Link: formula run	Run code, e.g. SU900110. SKUs made together share it; a trailing A marks a moved run. Same value as batch Oracle Type and ticket Run Code.	R03, R04
LineCode, ShiftName	Attribute	Where and when the run ran. A zero run with no shift is a ghost of a moved run.	R03, R04
ProductionStartDateTime, ProductionEndDateTime	Date	Run start and end. The agent orders a SKU's runs by start; the first one sets implied inventory.	R01, R02, R05

Column	Role	What it holds	Used by
NetUnitsOrdered	Quantity	What MIPS sent to production for this run. Negative = inventory already exceeds demand.	R01, R02, R05, R06, R08
ActualWrappedUnits	Quantity	Units the plant actually made.	R05, R06, R07, R17
MIPSOrderQty	Quantity	ODI's copy of the MIPS sales-order quantity.	R16
MIPSBeginningInventoryQty, MIPSInventoryAdjustmentQty	Quantity	Inventory MIPS recorded for the run.	R02
ExpectedEndingInventoryQty	Derived	ActualWrappedUnits - NetUnitsOrdered.	Evidence
TotalActualDamagedUnits, TotalPieces	Derived	Damaged units (0 throughout) and pieces made.	-

odi.batch (ODI)

One mixing record per formula run. It has no SKU column, so batch flags are keyed by formula and line.

Column	Role	What it holds	Used by
BatchEntryID	Key (synthetic)	BT-0001 to BT-0006.	-
Oracle Type	Link: formula run	Same value as OracleTypeCode on the finished-good runs.	R14, R17
Prod Line, Shift	Attribute	Where and when the batch was mixed.	R14
MIPS Sched Batches	Quantity	Batches MIPS scheduled.	R14
Act Batches Mixed	Quantity	Batches actually mixed.	R14
Theo Batch Pieces Qty	Quantity	Pieces per batch x batches mixed.	R14, R17
Final Piece Variance Qty	Quantity	Pieces over (+) or under (-) the theoretical count.	R17
Final Piece Variance %	Derived	Final Piece Variance Qty / Theo Batch Pieces Qty.	-

wms.order (WMS)

One customer order header with its fill and shipping progress (the Fill Rate report).

Column	Role	What it holds	Used by
Order Nbr	Key (native)	9900001 to 9900004, one per customer. WMS SHIP transactions carry it.	R12
Ship to Name	Link: customer	Same value as Cust Name on the item lines.	-
Order Type, Route Nbr	Attribute	TRANSFER_ORDER for the DCs; DSD_SALES_ORDER with a route number for the routes.	-
Order Qty, Allocated Qty	Quantity	Header totals. At seed time they equal the sum of the customer's item lines; after that nothing keeps them in step.	R13
Packed Qty, Loaded Qty, Shipped Qty	Quantity	Shipping progress. Shipped above 0 with Loaded 0 means no trailer load scan.	R12
Short Qty, % Allocated, % Loaded, % Shipped, Progress	Derived	Computed from the quantities above. Progress is GREEN when fully allocated.	-

wms.item_order (WMS)

Order quantity and allocation for one item and one customer (the Order Summary by Item report).

Column	Role	What it holds	Used by
Item Line ID	Key (synthetic)	WL-0001 to WL-0023. One row per SKU per customer.	-
Item Nbr	Link: SKU	Same value as MIPS FG Oracle Ref and ODI OracleFGNumber.	All SKU rules
Cust Name	Link: customer	Same value as Ship to Name on the order header.	R10 evidence
Sales Date	Date	The sales day. The Excel report prints it once in its header.	-
Original Ord Qty	Quantity	Quantity when the order was first taken.	-
Order Qty	Quantity	Units ordered now. Summed per SKU and compared with MIPS demand.	R09, R10, R13
Allocated Units Qty	Quantity	Units WMS has allocated to the customer.	R07, R10
Units per Container	Attribute	Units per tray in WMS.	R11
Stack and tray counts, Stacks / Trays Remaining, Completion%	Derived	21 trays per stack (an assumption). Completion% is GREEN when fully allocated.	-

wms.ticket (WMS)

One receipt ticket: finished goods received from a plant run into the warehouse or freezer.

Column	Role	What it holds	Used by
Ticket Nbr	Key (native)	TK-0001 to TK-0011, one per run that wrapped product.	R19
Item Nbr	Link: SKU	The SKU received.	R19
Run Code	Link: formula run	OracleTypeCode of the run that made the goods.	R19 evidence
Lot Nbr	Link: lot	L + receipt date (yymmdd) + last 4 characters of the run code. Repeated on the transaction and the Oracle posting.	-
Received Date Time	Date	Run end + 30 minutes.	-
Received Qty	Quantity	Units received. At seed time, the run's ActualWrappedUnits.	R19
Returned Qty	Quantity	Returns booked on the ticket (0 in the seed).	R19
Net Received Qty	Derived	Received Qty - Returned Qty.	R19
Location, Status	Attribute	DOCK and OPEN throughout the seed.	-

wms.transaction (WMS)

One inventory movement in WMS that Oracle must post.

Column	Role	What it holds	Used by
Txn Nbr	Key (native)	WT-0001 to WT-0011 in the seed; scenarios add WT-S5A, WT-S7A, WT-S8A to WT-S8F. Oracle's SourceReference points back to it.	R18
Txn Type	Attribute	RECEIPT (seed), SHIP (S8), RETURN_TO_PLANT (S5), FREEZER_OUT (S7).	R18, R19, returns
Txn Date Time	Date	When WMS booked it. Starts R18's 10-minute clock.	R18
Item Nbr	Link: SKU	The SKU moved.	R18, returns
Qty	Quantity	Units moved.	R18, R19, returns

Column	Role	What it holds	Used by
Ticket Nbr	Link: ticket	The ticket a receipt or return belongs to.	R19
Order Nbr	Link: order	The customer order a shipment belongs to.	R18 evidence
Lot Nbr, Reason Code, User	Attribute	Lot, reason (e.g. DAMAGED, FILL) and who booked it.	-

oracle.inventory_txn (ORACLE)

One Oracle Fusion inventory posting of a WMS transaction.

Column	Role	What it holds	Used by
TransactionId	Key (native)	OT-0001 to OT-0011 in the seed.	-
SourceReference	Link: WMS transaction	The Txn Nbr of the WMS transaction this posting records.	R18
ItemNumber	Link: SKU	The SKU posted.	-
TransactionTypeName	Attribute	WMS Receipt, WMS Shipment, WMS Return to Plant or WMS Freezer Issue.	-
TransactionQuantity	Quantity	Signed units: shipments and returns are negative.	-
TransactionDate, PostedDate	Date	In the seed, posted 2 minutes after the WMS receipt.	-
ProcessStatus	Attribute	POSTED or ERROR. Only POSTED counts as posted.	R18
ErrorMessage	Attribute	Oracle's reason for an ERROR.	R18 evidence
LotNumber, Subinventory, TransactionUnitOfMeasure	Attribute	Lot, FG and Each throughout.	-

Roles. *Key*: identifies the row (synthetic keys are added by the event contract). *Link*: holds the same value as a column in another table. *Quantity*: a number the rules compare. *Derived*: computed by the source system whenever the row changes (grey headers in the workbook).

5.10 How the tables connect

Eight kinds of value link the tables. The SKU is the main one: the agent adds up each system's rows per SKU and compares the totals. The others link single rows.

Link	Column	Same value in	What uses it
SKU	MIPS FG Oracle Ref	ODI OracleFGNumber; WMS Item Nbr (item lines, tickets, transactions); Oracle ItemNumber	Every SKU rule: each system's rows are added up per SKU and the totals compared
Legacy code	MIPS FG BUSS Code	ODI LegacyFGNumber	Nothing: carried, not matched
Formula run	ODI finished-good OracleTypeCode	ODI batch Oracle Type; WMS ticket Run Code	R04 finds the run a ghost moved to (a trailing A is ignored)
Customer	WMS item line Cust Name	WMS order Ship to Name	Rebuilding a header from its lines (simulator helper, R13 correction)
Ticket	WMS ticket Ticket Nbr	WMS transaction Ticket Nbr	R19
WMS transaction	WMS transaction Txn Nbr	Oracle SourceReference	R18
Customer order	WMS transaction Order Nbr	WMS order Order Nbr	Shipment detail (S8); no rule
Lot	WMS ticket Lot Nbr	WMS transaction Lot Nbr; Oracle LotNumber	Nothing yet

Before matching, the agent cleans every SKU value: it removes spaces, a trailing ".0" and leading zeros, so "900101 " and 900101 match. After that the match is exact. Rule R15 reports the padded BUSS values.

How the seed fills the links

The seed builds each table from the one before it, so the starting day is consistent wherever it should be:

- **ODI run → WMS ticket.** Each run that wrapped product gets a ticket: Received Qty = ActualWrappedUnits, Run Code = OracleTypeCode, received 30 minutes after the run ends.
- **Ticket → WMS transaction.** Each ticket gets a RECEIPT with the same Ticket Nbr, Qty and Lot Nbr.
- **WMS transaction → Oracle posting.** Each receipt gets a POSTED row whose SourceReference is the Txn Nbr, 2 minutes later.
- **WMS item lines → WMS order header.** Each customer's Order Qty and Allocated Qty are the sums of that customer's item lines.

Then each SKU gets one deliberate mismatch between MIPS, ODI and WMS (Section 7), and 900111 and 900112 stay clean.

5.11 What happens when a row changes

A change to one row never updates another table by itself. This is deliberate. At the plant, people and interfaces update each system separately, and an interface can lag, drop a change or never run. A simulation that kept the systems in step automatically could never show a mismatch, and finding mismatches is the agent's job.

Within the row. When a row changes, the source system recomputes its derived columns (Section 5.9). For example, a new ActualWrappedUnits also changes ExpectedEndingInventoryQty, and a new Returned Qty changes the ticket's Net Received Qty. wms.transaction and oracle.inventory_txn have no derived columns.

Across tables. Other tables change only in four ways:

- **Scenario scripts.** One step can change several tables at once. Section 9 lists every change.
- **Manual edits in the simulator.** The only cross-row helper is *Rebuild totals from item lines*, which sets a wms.order header to the sum of its item lines.
- **Approved corrections (write-back).** The worklist sends the fix to the simulator's stand-in for Oracle Integration, which changes the other system's row. For example, the R09 fix sets the MIPS sales-order line to match WMS.
- **The seed**, once, when it builds the starting day.

Example: scenario S6

Route 12 raises its SIM CINNAMON RAISIN order in WMS. MIPS is not refreshed.

Table	Row	What happens	Changed?
wms.item_order	WL-0023 (900112, Route 12)	Order Qty 900 → 1,100. Allocated stays 900, so Completion% (derived) turns YELLOW.	Yes
wms.order	9900003 (Route 12)	Order Qty 7,900 → 8,100. Short Qty (derived) is recomputed.	Yes: the scenario changes it too
mips.order_line	ML-0017 (900112)	Order Quantity stays 900.	No
odi.finished_good	FG-0014 (900112)	NetUnitsOrdered and MIPSOrderQty stay 900.	No

The agent matches the rows on SKU 900112: 1,100 ordered in WMS against 900 demanded in MIPS. It raises R09 (200 units more in WMS) and R10 (WMS allocation 200 short).

To see this live, open the worklist on the event feed and edit a row in the simulator (python -m recon_agent simulate). Only the edited row and its derived columns change, and the worklist raises or clears flags within about a second.

6. The reconciliation rules (R01-R19)

The engine checks the data with 19 rules. Each finding it raises is a **flag** whose ID is the rule code plus a key, for example `R05-900102`. Rules R01-R17 run on both the daily Excel extracts and the simulated event feed. **R18 and R19 run only on the event feed**, because only it carries WMS tickets, transactions and Oracle postings.

Rules work at different levels. **SKU** rules compare one SKU's MIPS demand, ODI production and WMS orders. **Run** and **batch** rules look at single ODI records. **Plant** rules raise one flag for the whole day. **Transaction** and **ticket** rules check WMS against Oracle in real time. R17 is statistical rather than a fixed condition (see anomaly.py).

6.1 Summary

Rule	Title	Requirement scenario	Scope	Severity	Usual owner	In dummy data
R01	Inventory applied to first order line only	1 - Multiple order lines for the same SKU	SKU	High if risk is 1,000 units or more, otherwise Medium	Production planner	Starting day, S1
R02	Inventory netted without audit trail	2 - Inventory on a zero-qty / blind line	SKU	Medium	Production planner	Starting day, S2
R03	Zero-quantity (blind) production line	2 - Inventory on a zero-qty / blind line	Run	Low	Production planner	Starting day, S2
R04	Production run moved; ghost run left behind	3 - Inventory does not follow a moved run	Run	Medium	Production planner	Starting day, S3
R05	Recovery run over-built (double-counted demand)	4 - Additional / recovery production runs	SKU	High if over-build is 1,000 units or more, otherwise Medium	Shift supervisor	Starting day, S4
R06	Excess production not carried to next demand	1/4 - Excess not applied to subsequent demand	SKU	Medium if surplus is 1,000 units or more, otherwise Low	Production planner	Starting day
R07	WMS filled demand from untracked inventory	7 - Freezer inventory remains after use	SKU	High if the gap is 500 units or more, otherwise Medium	Warehouse / WMS	Starting day, S5, S7
R08	Negative net order: inventory exceeds demand	2 - Inventory on a zero-qty / blind line	SKU	Medium	Production planner	Starting day
R09	MIPS vs WMS order quantity mismatch	6 - Screen refresh / stale data	SKU	Medium	IT / integration	Starting day, S6
R10	WMS allocation short of order	5 - Net quantity not reflected	SKU	Medium	Warehouse / WMS	Starting day, S6
R11	Units-per-container master data conflict	Data harmonization	SKU	Medium	Master data	Starting day
R12	Shipped without trailer load scan	8 - Missing downstream transactions	Plant	High	Warehouse / WMS	Starting day
R13	Cross-report total mismatch	8 - Missing downstream transactions	Plant / item	Medium	IT / integration	Not seeded
R14	Batches mixed differ from MIPS schedule	Order-to-batch traceability	Batch	Low	Shift supervisor	Starting day
R15	SKU key formatting differs by source system	Data harmonization	Plant	Low	IT / integration	Starting day

Rule	Title	Requirement scenario	Scope	Severity	Usual owner	In dummy data
R16	ODI MIPS order qty disagrees with MIPS sales orders	8 - Missing downstream transactions	SKU	Medium	IT / integration	Not seeded
R17	Statistical outlier - possible data-entry error	Unusually large quantity	Run / batch	Medium; High if the robust z-score is 7 or more	Shift supervisor	Not seeded
R18	WMS transaction not posted in Oracle within 10 minutes	8 - Missing downstream transactions	Transaction	High	IT / integration *	S8
R19	Return to plant did not reduce received quantity	5 - Net quantity not reflected	Ticket	High if the gap is 500 units or more, otherwise Medium	Warehouse / WMS *	S5

* The user guide lists owners for R01-R17 only; the owners of R18 and R19 are inferred from the systems they check. The requirement scenario numbers refer to the eight scenarios in the requirement brief (docs/reference/WMS_AI_Agent_Requirement.docx).

6.2 Quantities the rules use

SKU rules work on a reconciled position per SKU, built in position.py:

Quantity	Definition
Demand	Sum of the SKU's MIPS order quantities, all lines.
Current-day demand	Demand less lines moved to the prior sales day (Order Moved = TRUE).
Sales-order demand	Demand on BUSS lines only (OCloud supply-plan lines excluded).
Implied inventory	Inventory MIPS netted out before production: demand less what was sent to the first active run (or less total net ordered when there is one run). Recovery runs do not reduce it.
Recorded inventory	MIPSBeginningInventoryQty + MIPSInventoryAdjustmentQty over the SKU's runs.
Wrapped	Sum of ActualWrappedUnits over the SKU's runs.
Consolidated requirement	Demand less implied inventory: what the plant should make for the SKU.
Net requirement	Demand - implied inventory - wrapped + returns. Positive: still to produce (or filled from unseen stock). Negative: surplus to carry forward.
Active run	An ODI run with a non-zero ordered or wrapped quantity.

Thresholds (recon_agent/config.py)

Setting	Value	Used by
QTY_TOLERANCE_UNITS	0 units	R09, R13, R16, R19: quantities must match exactly
CARRYOVER_MIN_UNITS	50 units	R06: ignore trivial end-of-run excess
OVERPRODUCTION_MIN_UNITS	100 units	R05: smallest over-build worth flagging
R07 minimum gap	20 units	R07 (fixed in the rule)
ROBUST_Z_THRESHOLD	3.5	R17: outlier cut-off (High at 7.0)
ORACLE_POSTING_SLA_MINUTES	10 plant minutes	R18: posting window
DEFAULT_UNIT_PRICE	$3.99	Dollar value when a SKU has no MIPS price (R13, unknown SKUs)

6.3 Ranking and dollar exposure

Each flag's dollar value is **units x retail price** of the SKU. The worklist ranks flags by:

```
Priority = (dollar value + base value) x severity weight x rule confidence
```

- **Severity weight:** High 3, Medium 2, Low 1.
- **Base value** gives flags without dollars a place in the order: 50 by default; R14 500; R04 1,000; R13 (report totals) 1,000; R17 (piece variance) 1,500; R18 2,000; R12 5,000.
- **Rule confidence** starts at 1.0 and goes down when planners reject a rule's flags. Decisions made in the simulation never change the confidence learned from real data.
- **Dollar exposure** (the worklist KPI) adds the dollar values of R01, R05, R06, R07, R08, R09, R10 and R19 only.
- **Carry-forward:** accepting an R05, R06 or R08 flag shows how the surplus moves to the next sales day.

6.4 Rule by rule

Each card shows what the rule detects, the exact condition in the code, what the flag measures, and where the dummy data shows it.

R01 · Inventory applied to first order line only

Requirement scenario	1 - Multiple order lines for the same SKU
Detects	MIPS applies carried-in inventory to the first order line of a SKU instead of the SKU's total, so the remaining lines ask the plant to make more than is needed.
Triggers when	The SKU has two or more MIPS lines **and** the first-line over-production risk is above 0. Risk = (demand - min(implied inventory, first line qty)) - (demand - implied inventory).
Measures	The over-production risk in units.
Severity	High if risk is 1,000 units or more, otherwise Medium
Systems / owner	MIPS, ODI · Production planner · area: Inventory Allocation
Recommended action	Consolidate the SKU's lines before applying inventory; produce to the SKU-level requirement.
Write-back fix	None (needs a planner).
In the dummy data	Starting day: R01-900101 (High, 1,000 units, $4,690) Scenarios: S1 raises R01-900111 (Medium, 500 units, $1,995)

R02 · Inventory netted without audit trail

Requirement scenario	2 - Inventory on a zero-qty / blind line
Detects	MIPS sent less to production than the demand, so it netted inventory out, but no beginning inventory or adjustment is recorded. Nobody can trace where that inventory is.
Triggers when	Implied inventory is above 0, MIPSBeginningInventoryQty + MIPSInventoryAdjustmentQty is 0, R01 does not apply, and no run has a negative net order.
Measures	The implied inventory in units.
Severity	Medium
Systems / owner	MIPS, ODI · Production planner · area: Inventory Synchronization
Recommended action	Find the schedule line holding the inventory (often a zero-quantity or planning line) and record it as beginning inventory.
Write-back fix	None by design: recording inventory blindly could hide a real shortfall.
In the dummy data	Starting day: R02-900103 (Medium, 800 units, $3,192) Scenarios: S2 raises R02-900112 (Medium, 200 units, $1,058)

R03 · Zero-quantity (blind) production line

Requirement scenario	2 - Inventory on a zero-qty / blind line
Detects	A production line sits in a live run with zero ordered and zero wrapped. Inventory applied to it never reduces real demand.
Triggers when	An ODI run has NetUnitsOrdered = 0 and ActualWrappedUnits = 0, and it is not the ghost of a moved run (see R04).
Measures	None (0 units; it is a control flag).
Severity	Low
Systems / owner	MIPS, ODI · Production planner · area: Order-to-Batch Traceability
Recommended action	Move any allocation to the active line and close the blind line.
Write-back fix	None.
In the dummy data	Starting day: R03-SU900130-900104 (Low, 0 units) Scenarios: S2 raises R03-SU900220-900112 (Low, 0 units)

R04 · Production run moved; ghost run left behind

Requirement scenario	3 - Inventory does not follow a moved run
Detects	A run was moved to another line and ODI left a zero-quantity copy on the old line. Inventory and WMS work orders may still point at the ghost.
Triggers when	A zero-ordered, zero-wrapped run has **no shift**, and the same formula (ignoring a trailing "A") produced output on a different line.
Measures	None (0 units).
Severity	Medium
Systems / owner	MIPS, ODI · Production planner · area: Inventory Synchronization
Recommended action	Confirm inventory and WMS allocations moved with the run, then remove the ghost entry.
Write-back fix	None.
In the dummy data	Starting day: R04-SU900150-900105 (Medium, 0 units) Scenarios: S3 raises R04-SU900210-900111 (Medium, 0 units)

R05 · Recovery run over-built (double-counted demand)

Requirement scenario	4 - Additional / recovery production runs
Detects	A recovery run was sized to the full demand instead of the remaining shortfall, so the SKU is over-built.
Triggers when	The SKU has two or more active runs **and** either over-build (wrapped - consolidated requirement) or re-order (ODI net ordered - demand) is at least 100 units.
Measures	The over-build in units (never below 0).
Severity	High if over-build is 1,000 units or more, otherwise Medium
Systems / owner	MIPS, ODI · Shift supervisor · area: Inventory Allocation
Recommended action	Carry the surplus to the next sales day (or the freezer); size recovery runs to the shortfall only.
Write-back fix	None by design: which day takes the surplus is a planner decision. Accepting the flag shows the carry-forward.
In the dummy data	Starting day: R05-900102 (High, 1,200 units, $5,628) Scenarios: S4 raises R05-900112 (Medium, 800 units, $4,232)

R06 · Excess production not carried to next demand

Requirement scenario	1/4 - Excess not applied to subsequent demand
Detects	A single run made more than the demand, and the surplus is not carried forward as next-day beginning inventory.
Triggers when	The SKU has one active run, surplus (= -net requirement) is at least 50 units, and no run has a negative net order.
Measures	The surplus in units.
Severity	Medium if surplus is 1,000 units or more, otherwise Low
Systems / owner	ODI, MIPS, WMS · Production planner · area: Inventory Allocation
Recommended action	Carry the surplus to the next sales day and reduce the next production requirement by the same amount.
Write-back fix	None by design (planner decision). Accepting the flag shows the carry-forward.
In the dummy data	Starting day: R06-900108 (Low, 400 units, $1,796)

R07 · WMS filled demand from untracked inventory

Requirement scenario	7 - Freezer inventory remains after use
Detects	WMS filled the order in full although production plus applied inventory falls short: the difference came from stock MIPS cannot see, usually the freezer or prior-day tickets.
Triggers when	Net requirement (demand - implied inventory - wrapped + returns) is at least 20 units **and** WMS allocated at least the current-day demand (moved lines excluded).
Measures	The gap in units.
Severity	High if the gap is 500 units or more, otherwise Medium
Systems / owner	WMS, MIPS, ODI · Warehouse / WMS · area: Freezer Inventory Management
Recommended action	Confirm the freezer or ticket draw in WMS and deduct it in MIPS, or freezer balances stay overstated.
Write-back fix	None by design: which freezer lot was used needs a person.
In the dummy data	Starting day: R07-900106 (High, 700 units, $2,513) Scenarios: S5 raises R07-900111 (Medium, 120 units, $479); S7 raises R07-900111 (Medium, 300 units, $1,197)

R08 · Negative net order: inventory exceeds demand

Requirement scenario	2 - Inventory on a zero-qty / blind line
Detects	Inventory already exceeds demand, so ODI shows a negative net order and no production is needed.
Triggers when	Any ODI run for the SKU has NetUnitsOrdered below 0.
Measures	The excess in units (sum of the negative net orders).
Severity	Medium
Systems / owner	MIPS, ODI, WMS · Production planner · area: Inventory Synchronization
Recommended action	Remove the run from the schedule and verify the surplus exists in WMS before carrying it forward.
Write-back fix	None. Accepting the flag shows the carry-forward.
In the dummy data	Starting day: R08-900110 (Medium, 300 units, $1,437)

R09 · MIPS vs WMS order quantity mismatch

Requirement scenario	6 - Screen refresh / stale data
Detects	MIPS and WMS disagree on how much the customers ordered, usually because the order changed after MIPS last synced.
Triggers when	WMS order quantity differs from MIPS current-day demand (moved lines excluded). Tolerance: 0 units.
Measures	The difference in units (positive when WMS carries more).
Severity	Medium
Systems / owner	MIPS, WMS · IT / integration · area: Production Visibility
Recommended action	Confirm the WMS order and update the MIPS requirement, or refresh MIPS if MIPS carries more.
Write-back fix	MIPS follows WMS: the SKU's largest sales-order line takes the difference.
In the dummy data	Starting day: R09-900107 (Medium, 200 units, $718) Scenarios: S6 raises R09-900112 (Medium, 200 units, $1,058)

R10 · WMS allocation short of order

Requirement scenario	5 - Net quantity not reflected
Detects	WMS allocated less than was ordered, often because returned or damaged tickets were not netted, or a workstation shows stale data.
Triggers when	WMS allocated units are below WMS order quantity for the SKU. The flag names the short customers.
Measures	The shortfall in units.
Severity	Medium
Systems / owner	WMS, ODI · Warehouse / WMS · area: Warehouse Traceability
Recommended action	Check returned or damaged tickets and stale workstation data before creating a new work order.
Write-back fix	None.
In the dummy data	Starting day: R10-900107 (Medium, 200 units, $718) Scenarios: S6 raises R10-900112 (Medium, 200 units, $1,058)

R11 · Units-per-container master data conflict

Requirement scenario	Data harmonization
Detects	MIPS and WMS use a different number of units per tray for the same SKU, so stack and tray counts cannot tie out.
Triggers when	MIPS units per container (the most common value on container code 01 lines) differs from WMS units per container.
Measures	None (0 units).
Severity	Medium
Systems / owner	MIPS, WMS · Master data · area: Data Flow Transparency
Recommended action	Align the item master; Oracle is the system of record.
Write-back fix	None.
In the dummy data	Starting day: R11-900109 (Medium, 0 units)

R12 · Shipped without trailer load scan

Requirement scenario	8 - Missing downstream transactions
Detects	Orders show as shipped but no trailer load scan was recorded, so shipments cannot be traced ticket to trailer.
Triggers when	Any WMS order has Shipped Qty above 0 and Loaded Qty = 0. One plant-level flag covers all such orders.
Measures	The shipped units on those orders ($0: no SKU price applies).
Severity	High
Systems / owner	WMS, Oracle · Warehouse / WMS · area: Warehouse Traceability
Recommended action	Confirm the ship transactions reached Oracle and reinstate load scanning.
Write-back fix	None.
In the dummy data	Starting day: R12-PLANT (High, 7,900 units)

R13 · Cross-report total mismatch

Requirement scenario	8 - Missing downstream transactions
Detects	Two WMS views of the same orders do not add up, so an order line is missing from one of them or changed between runs.
Triggers when	The Fill Rate total (event feed: the sum of order headers) differs from the sum of item-line order quantities. With Excel extracts, also any item whose printed subtotal differs from the sum of its lines.
Measures	The difference in units, valued at the $3.99 default price.
Severity	Medium
Systems / owner	WMS · IT / integration · area: Integration Architecture
Recommended action	Reconcile at order-line level before financial close.
Write-back fix	Rebuild the most-mismatched WMS order header from its item lines.
In the dummy data	**Not seeded.** In the simulator, change a WMS order header's Order Qty so it no longer equals its item lines.

R14 · Batches mixed differ from MIPS schedule

Requirement scenario	Order-to-batch traceability
Detects	More or fewer dough batches were mixed than MIPS scheduled, so output will differ from plan.
Triggers when	An ODI batch has MIPS Sched Batches above 0 and Act Batches Mixed different from it.
Measures	None (0 units). The evidence estimates extra pieces = theoretical pieces / batches mixed x difference.
Severity	Low
Systems / owner	MIPS, ODI · Shift supervisor · area: Order-to-Batch Traceability
Recommended action	Record the reason (recovery or rework) so the extra output is expected in WMS.
Write-back fix	None.
In the dummy data	Starting day: R14-SU900121-Line 2 (Low, 0 units)

R15 · SKU key formatting differs by source system

Requirement scenario	Data harmonization
Detects	SKU codes carry extra spaces in one source, so exact-match joins to WMS and Oracle silently fail.
Triggers when	Any MIPS FG Oracle Ref has leading or trailing spaces. One plant-level flag; the agent normalises keys in the meantime.
Measures	None (0 units).
Severity	Low
Systems / owner	MIPS · IT / integration · area: Data Flow Transparency
Recommended action	Trim the key at the interface.
Write-back fix	None.
In the dummy data	Starting day: R15-MIPS-FG-REF (Low, 0 units)

R16 · ODI MIPS order qty disagrees with MIPS sales orders

Requirement scenario	8 - Missing downstream transactions
Detects	ODI's copy of the MIPS sales-order quantity is out of date, so an order transaction may have been dropped between MIPS and ODI.
Triggers when	The SKU has an active run and MIPS lines, and the sum of MIPSOrderQty across its ODI runs differs from its MIPS sales-order (BUSS) demand. Tolerance: 0 units.
Measures	The difference in units.
Severity	Medium
Systems / owner	MIPS, ODI · IT / integration · area: Integration Architecture
Recommended action	Re-run the MIPS to ODI interface for the SKU.
Write-back fix	Re-send the MIPS sales-order quantity to the SKU's first ODI run.
In the dummy data	**Not seeded.** In the simulator, change an ODI run's MIPSOrderQty (for example FG-0012 or FG-0013).

R17 · Statistical outlier - possible data-entry error

Requirement scenario	Unusually large quantity
Detects	A value is far from the plant's normal range and may be a keying error.
Triggers when	Robust z-score (median / MAD) of 3.5 or more, among at least 8 samples, on (a) wrap overrun per run = (wrapped - ordered) / ordered, or (b) final piece variance per batch.
Measures	(a) wrapped - ordered; (b) none.
Severity	Medium; High if the robust z-score is 7 or more
Systems / owner	ODI · Shift supervisor · area: Production Visibility
Recommended action	Verify the wrapped count, ordered quantity or scrap entries before any surplus is carried forward.
Write-back fix	None.
In the dummy data	**Not seeded.** In the simulator, set one run's ActualWrappedUnits far above its NetUnitsOrdered (for example FG-0011 to 5,400); this also raises R06 for the surplus. The batch check cannot fire on the dummy data: it has only 6 batches, fewer than the 8 samples needed.

R18 · WMS transaction not posted in Oracle within 10 minutes

Requirement scenario	8 - Missing downstream transactions
Detects	A WMS inventory transaction has not posted in Oracle Fusion, so Oracle inventory and finance are understated. Real-time only.
Triggers when	A WMS transaction has no POSTED Oracle transaction with SourceReference = its Txn Nbr, and it is at least 10 plant minutes old. Covers both missing and ERROR postings.
Measures	The transaction quantity.
Severity	High
Systems / owner	WMS, Oracle · IT / integration · area: Integration Architecture
Recommended action	Resubmit through the integration and check the interface error queue.
Write-back fix	Resubmit the posting (repost an ERROR, or insert the missing one).
In the dummy data	Scenarios: S8 raises R18-WT-S8A (High, 3,000 units, $14,070); S8 raises R18-WT-S8F (High, 1,200 units, $5,748)

R19 · Return to plant did not reduce received quantity

Requirement scenario	5 - Net quantity not reflected
Detects	Units returned to the plant were not subtracted from the receipt ticket, so WMS can still allocate stock that is back at the plant. Real-time only.
Triggers when	The RETURN_TO_PLANT transactions booked against a ticket exceed that ticket's Returned Qty. Tolerance: 0 units.
Measures	The unbooked returned units.
Severity	High if the gap is 500 units or more, otherwise Medium
Systems / owner	WMS, MIPS · Warehouse / WMS · area: Warehouse Traceability
Recommended action	Reduce the ticket's net received quantity by the returned units.
Write-back fix	Book the returned quantity on the WMS ticket.
In the dummy data	Scenarios: S5 raises R19-TK-0010 (Medium, 120 units, $479)

7. The designed problems, step by step

Each row below shows the data that creates a flag and the arithmetic the engine applies. Units and dollar values are the engine's own output on the starting day. Section 6 defines each rule.

SKU / scope	Data that creates it	Flag	Units	$
900101	MIPS lines 1,300 (moved, listed first) + 2,400 + 7,600 = 11,300 demand. ODI ordered 9,000, so 2,300 units of inventory were applied. Netting it against the first line only asks for 10,000 instead of the SKU-level 9,000.	R01-900101	1,000	$4,690
900102	Demand 6,000. Run 1 ordered 6,000, wrapped 5,200 (800 short). The recovery run ordered the full 2,000. Wrapped total 7,200: 1,200 surplus.	R05-900102	1,200	$5,628
900102 batch	Recovery formula SU900121 on Line 2 mixed 4 batches against 3 scheduled (+1, about 500 pieces).	R14-SU900121-Line 2	-	-
900103	Demand 5,000; ODI sent 4,200 to production, so 800 units of inventory were netted, but MIPSBeginningInventoryQty is 0. Nothing records where the 800 came from.	R02-900103	800	$3,192
900104	A zero-quantity MIPS line and a zero ordered / zero wrapped run that still has a shift, inside the live formula SU900130.	R03-SU900130-900104	-	-
900105	Run SU900150 moved from Line 1 to Line 2 (SU900150A, 3,000 wrapped). A zero copy with no shift stays on Line 1.	R04-SU900150-900105	-	-
900106	Demand 4,000; wrapped 3,300. WMS allocated all 4,000 (2,400 North + 1,600 South): 700 came from stock MIPS cannot see.	R07-900106	700	$2,513
900107	MIPS demand 3,500 (2,000 + 1,500). WMS orders total 3,700 because Route 27 raised 800 to 1,000 after MIPS last synced (Thu 20:31).	R09-900107	200	$718
900107	Route 27 is allocated 800 of its 1,000.	R10-900107	200	$718
900108	Demand 2,500; wrapped 2,900. The 400 surplus is not carried to the next sales day.	R06-900108	400	$1,796
900109	10 units per container in MIPS, 12 in WMS: stacks and trays cannot tie out.	R11-900109	-	-
900110	Demand 1,200; MIPS beginning inventory 1,500. ODI shows a net order of -300 with nothing wrapped.	R08-900110	300	$1,437
Plant	Order 9900003 (Route 12) shows 7,900 shipped with 0 loaded: no trailer load scan.	R12-PLANT	7,900	-
Plant	7 of 17 MIPS lines (all BUSS lines) pad the FG Oracle Ref with a trailing space.	R15-MIPS-FG-REF	-	-
900111, 900112	Every number ties out across MIPS, ODI and WMS.	none	-	-

8. What the engine reports on the starting day

Run on the seeded data (AI explanations off), the engine reconciles 12 SKUs and raises **14 flags**: 4 High, 6 Medium and 4 Low. The worklist shows them on 13 rows (10 SKUs and 3 plant-level rows).

Indicator	Value
SKUs reconciled / clean	12 / 2
MIPS lines / ODI runs / WMS orders	17 / 14 / 4
Total demand (MIPS)	43,200 units
Total wrapped (ODI)	39,800 units
Total allocated (WMS)	41,900 units
Dollar exposure (R01, R05-R10, R19)	$17,500
Over-production risk (R01, R05)	2,200 units
Carry-over (R06, R08)	700 units
Untracked supply (R07)	700 units

Flag ID	Rule title	Severity	Owner area	Units	$
R05-900102	Recovery run over-built (double-counted demand)	High	Inventory Allocation	1,200	$5,628
R12-PLANT	Shipped without trailer load scan	High	Warehouse Traceability	7,900	-
R01-900101	Inventory applied to first order line only	High	Inventory Allocation	1,000	$4,690
R07-900106	WMS filled demand from untracked inventory	High	Freezer Inventory Management	700	$2,513
R02-900103	Inventory netted without audit trail	Medium	Inventory Synchronization	800	$3,192
R08-900110	Negative net order: inventory exceeds demand	Medium	Inventory Synchronization	300	$1,437
R04-SU900150-900105	Production run moved; ghost run left behind	Medium	Inventory Synchronization	-	-
R06-900108	Excess production not carried to next demand	Low	Inventory Allocation	400	$1,796
R09-900107	MIPS vs WMS order quantity mismatch	Medium	Production Visibility	200	$718
R10-900107	WMS allocation short of order	Medium	Warehouse Traceability	200	$718
R14-SU900121-Line 2	Batches mixed differ from MIPS schedule	Low	Order-to-Batch Traceability	-	-
R11-900109	Units-per-container master data conflict	Medium	Data Flow Transparency	-	-
R03-SU900130-900104	Zero-quantity (blind) production line	Low	Order-to-Batch Traceability	-	-
R15-MIPS-FG-REF	SKU key formatting differs by source system	Low	Data Flow Transparency	-	-

Flags appear in the engine's priority order. Severity can depend on size: R05 is High at 1,000 units or more, and R06 at 400 units is Low. The dollar value is units x retail price.

9. The scenario library (S1-S8)

Each scenario plays one of the eight requirement scenarios on top of the untouched starting day. Changes go through the normal producer path, so derived columns are recomputed and any active integration fault applies. The flags below were verified by running each scenario on a fresh seed for this document; none of the 14 starting flags is cleared by any scenario.

S1 · Inventory applied to the first order line only

Requirement scenario 1. MIPS nets 2,000 units of carried-in inventory for SIM BUTTER TOP, but applies them against the first of its two order lines only. The run is cut to 2,000.

Steps

1. MIPS sends 2,000 to production instead of 4,000; the run wraps 2,000.

Step	Entity	Op	Row	Change
1	odi.finished_good	U	FG-0013	NetUnitsOrdered: 4000 → 2000; ActualWrappedUnits: 4000 → 2000

Flag raised	Severity	Units	$	Why
R01-900111	Medium	500	$1,995	inventory covers only the first 1,500-unit line: 500 units of over-production risk

S2 · Inventory parked on a zero-quantity line

Requirement scenario 2. MIPS nets 200 units of inventory for SIM CINNAMON RAISIN through a new zero-quantity planning line, and ODI schedules that line as a blind run. Nothing records the inventory.

Steps

1. MIPS adds a zero-quantity supply-plan line and nets 200 units; the run is cut to 700.
2. ODI schedules the zero line on Line 1 as its own run.

Step	Entity	Op	Row	Change
1	mips.order_line	I	ML-S2A	new: FG Oracle Ref=900112, Order Quantity=0, Source System=OCloud Region2
1	odi.finished_good	U	FG-0014	NetUnitsOrdered: 900 → 700; ActualWrappedUnits: 900 → 700
2	odi.finished_good	I	FG-S2A	new: OracleFGNumber=900112, OracleTypeCode=SU900220, LineCode=Line 1, ShiftName=Shift 2, NetUnitsOrdered=0, ActualWrappedUnits=0

Flag raised	Severity	Units	$	Why
R02-900112	Medium	200	$1,058	200 units netted with no beginning inventory recorded
R03-SU900220-900112	Low	-	-	a zero-quantity run with a shift: a blind line

S3 · Production run moved to another line

Requirement scenario 3. SIM BUTTER TOP's run moves from Line 2 to Line 1 (formula SU900210A). ODI leaves a zero copy with no shift on Line 2.

Steps

1. The run is re-created on Line 1; the Line 2 entry is zeroed and loses its shift.

Step	Entity	Op	Row	Change
1	odi.finished_good	I	FG-S3A	new: OracleFGNumber=900111, OracleTypeCode=SU900210A, LineCode=Line 1, ShiftName=Shift 1, NetUnitsOrdered=4000, ActualWrappedUnits=4000
1	odi.finished_good	U	FG-0013	ShiftName: Shift 1 → None; NetUnitsOrdered: 4000 → 0; ActualWrappedUnits: 4000 → 0; MIPSOrderQty: 1500 → 0

Flag raised	Severity	Units	$	Why
R04-SU900210-900111	Medium	-	-	the zero copy left on Line 2 is a ghost of the moved run

S4 · Recovery run double-counts demand

Requirement scenario 4. SIM CINNAMON RAISIN's run wraps 100 short. WMS still fills from stock, then a recovery run is ordered for the full 900 instead of the 100 missing.

Steps

1. The run wraps 800 of 900.
2. Ninety minutes later a recovery run on Line 1 makes the full 900 again. *(after 90m of plant time)*

Step	Entity	Op	Row	Change
1	odi.finished_good	U	FG-0014	ActualWrappedUnits: 900 → 800
2	odi.finished_good	I	FG-S4A	new: OracleFGNumber=900112, OracleTypeCode=SU900221, LineCode=Line 1, ShiftName=Shift 3, NetUnitsOrdered=900, ActualWrappedUnits=900

Flag raised	Severity	Units	$	Why
R05-900112	Medium	800	$4,232	the recovery run over-builds by 800

S5 · Return to plant does not reduce the received quantity

Requirement scenario 5. WMS returns 120 damaged units of SIM BUTTER TOP to the plant against ticket TK-0010. Oracle posts the return, but the ticket keeps its full received quantity.

Steps

1. WMS books the return; Oracle posts it.

Step	Entity	Op	Row	Change
1	wms.transaction	I	WT-S5A	new: Txn Type=RETURN_TO_PLANT, Item Nbr=900111, Qty=120, Ticket Nbr=TK-0010, Reason Code=DAMAGED
1	oracle.inventory_txn	I	OT-S5A	new: SourceReference=WT-S5A, TransactionTypeName=WMS Return to Plant, TransactionQuantity=-120, ProcessStatus=POSTED

Flag raised	Severity	Units	$	Why
R07-900111	Medium	120	$479	WMS allocated 4,000 but only 3,880 usable units were made
R19-TK-0010	Medium	120	$479	the ticket still shows 4,000 received, not 3,880

S6 · Order changed after MIPS last synced (stale screen)

Requirement scenario 6. Route 12 raises its SIM CINNAMON RAISIN order from 900 to 1,100 in WMS. MIPS has not refreshed, so its screen still plans 900.

Steps

1. WMS takes the bigger order; the order header follows.

Step	Entity	Op	Row	Change
1	wms.item_order	U	WL-0023	Order Qty: 900 → 1100
1	wms.order	U	9900003	Order Qty: 7900 → 8100

Flag raised	Severity	Units	$	Why
R09-900112	Medium	200	$1,058	WMS carries 200 more than MIPS scheduled
R10-900112	Medium	200	$1,058	WMS is 200 short against the bigger order

S7 · Freezer inventory not deducted

Requirement scenario 7. Route 27 needs 300 more SIM BUTTER TOP. WMS fills them from the freezer and MIPS raises the plan, but nothing deducts the freezer stock from what MIPS believes was made.

Steps

1. MIPS plans 300 more and sends them to ODI, but the plant makes nothing extra: WMS allocates them from the freezer and posts the move.

Step	Entity	Op	Row	Change
1	mips.order_line	U	ML-0016	Order Quantity: 2500 → 2800
1	odi.finished_good	U	FG-0013	NetUnitsOrdered: 4000 → 4300
1	wms.item_order	U	WL-0022	Order Qty: 1000 → 1300; Allocated Units Qty: 1000 → 1300
1	wms.order	U	9900004	Order Qty: 4200 → 4500; Allocated Qty: 4000 → 4300
1	wms.transaction	I	WT-S7A	new: Txn Type=FREEZER_OUT, Item Nbr=900111, Qty=300, Reason Code=FILL
1	oracle.inventory_txn	I	OT-S7A	new: SourceReference=WT-S7A, TransactionTypeName=WMS Freezer Issue, TransactionQuantity=300, ProcessStatus=POSTED

Flag raised	Severity	Units	$	Why
R07-900111	Medium	300	$1,197	300 units shipped from stock MIPS cannot see

S8 · WMS transactions missing in Oracle

Requirement scenario 8. The SIM DC SOUTH trailer ships. Oracle posts four of the six item shipments, rejects one, and never receives the sixth. After 10 plant minutes both are overdue.

Steps

1. WMS ships order 9900002, one transaction per item.
2. Oracle posts four, rejects 900110's (lot not found), and never sees 900101's. *(after 2m of plant time)*
3. Ten plant minutes pass. *(after 9m of plant time)*

Step	Entity	Op	Row	Change
1	wms.order	U	9900002	Loaded Qty: 0 → 11800; Shipped Qty: 0 → 11800
1	wms.transaction	I	WT-S8A	new: Txn Type=SHIP, Item Nbr=900101, Qty=3000, Order Nbr=9900002
1	wms.transaction	I	WT-S8B	new: Txn Type=SHIP, Item Nbr=900102, Qty=2000, Order Nbr=9900002
1	wms.transaction	I	WT-S8C	new: Txn Type=SHIP, Item Nbr=900103, Qty=2500, Order Nbr=9900002
1	wms.transaction	I	WT-S8D	new: Txn Type=SHIP, Item Nbr=900106, Qty=1600, Order Nbr=9900002
1	wms.transaction	I	WT-S8E	new: Txn Type=SHIP, Item Nbr=900108, Qty=1500, Order Nbr=9900002
1	wms.transaction	I	WT-S8F	new: Txn Type=SHIP, Item Nbr=900110, Qty=1200, Order Nbr=9900002

Step	Entity	Op	Row	Change
2	oracle.inventory_txn	I	OT-S8B	new: SourceReference=WT-S8B, TransactionTypeName=WMS Shipment, TransactionQuantity=-2000, ProcessStatus=POSTED
2	oracle.inventory_txn	I	OT-S8C	new: SourceReference=WT-S8C, TransactionTypeName=WMS Shipment, TransactionQuantity=-2500, ProcessStatus=POSTED
2	oracle.inventory_txn	I	OT-S8D	new: SourceReference=WT-S8D, TransactionTypeName=WMS Shipment, TransactionQuantity=-1600, ProcessStatus=POSTED
2	oracle.inventory_txn	I	OT-S8E	new: SourceReference=WT-S8E, TransactionTypeName=WMS Shipment, TransactionQuantity=-1500, ProcessStatus=POSTED
2	oracle.inventory_txn	I	OT-S8F	new: SourceReference=WT-S8F, TransactionTypeName=WMS Shipment, TransactionQuantity=-1200, ProcessStatus=ERROR, ErrorMessage=Lot not found in subinventory FG

Flag raised	Severity	Units	$	Why
R18-WT-S8A	High	3,000	$14,070	Oracle never received the 900101 shipment
R18-WT-S8F	High	1,200	$5,748	Oracle rejected the 900110 shipment

S4 shows a transient R07 between its two steps (WMS has filled from stock before the recovery run lands). The final state raises R05 only. S5's R07 appears because WMS allocated 4,000 while only 3,880 usable units remain.

10. Conventions copied from the client extracts

The dummy data reproduces the oddities of the real files on purpose, so the simulation exercises the same ingest and matching logic as the batch mode:

Convention	How it appears in the dummy data
BUSS pads the FG Oracle Ref	All 7 BUSS lines carry "9001xx " with a trailing space; OCloud lines do not.
Moved lines keep the schedule date	ML-0001: Schedule Date = sales day, but Offset / BUSS / Product Sales Date = prior day, Order Moved = TRUE.
Runs share an OracleTypeCode per formula	SU900130 covers 900103 and 900104; SU900160 covers 900106 and 900107.
A moved run leaves a zero copy with no shift	FG-0006 (SU900150, Line 1) beside FG-0007 (SU900150A, Line 2).
Recovery runs get their own formula suffix	SU900121 for 900102; S4 adds SU900221 for 900112.
BUSS lines have no production date	Production Date is empty on BUSS lines; OCloud lines carry sales day - 1.
WMS % Loaded / % Shipped are fractions	% Allocated is 0-100, % Loaded and % Shipped are 0-1, as in the Fill Rate report.
Synthetic keys	Order Line ID, FGEntryID, BatchEntryID and Item Line ID do not exist in the extracts (contract assumptions A1-A4).

11. Guide to the Excel workbook

Sheet	What it contains
Overview	Key facts, a linked index of sheets with live row counts, and the header colour legend.
SKU Master	The 12 SKUs: BUSS code, description, price, units per tray, designed problem, expected flags, scenarios.
Customers	The 4 customers: order number, type, route and state at clock start.
MIPS Order Lines ... Oracle Inventory Postings	One sheet per source table (8), every contract column in contract order, filters and frozen headers, plus a yellow 'Why This Row Exists' column.
Seed Flags	The 14 starting-day flags: rule, severity, owner, units, price, dollars, designed reason, engine recommendation; SUM totals.
Scenarios	S1-S8: story, steps, number of changes, flags raised (verified) and why.
Scenario Changes	All 30 changes the scenarios publish: plant time, entity, operation, row, before → after values.
Data Dictionary	Every column of every table: type, key, required and derived flags.

Header colours

Colour	Meaning
MIPS	MIPS (planning) columns
ODI	ODI / Oracle production columns
WMS	WMS columns
ORACLE	Oracle Fusion columns
Grey	Derived column: computed by the source system, not entered (cell comment explains)
Yellow	Documentation-only column: not part of any source system

Header cells carry comments for row keys (and whether they are synthetic) and for derived columns. The Overview row counts are formulas, so they update if rows are added or removed.

12. Limitations

- **Small by design.** 12 SKUs and 4 customers, against roughly 100 findings on a real plant day. It is built to show each rule clearly, not to test volume or performance.
- **One problem per SKU.** Real SKUs often carry several overlapping problems; only 900102 (R05, R14) and 900107 (R09, R10) carry two here.
- **No damage, scrap or weights.** Damaged-unit and weight columns are zero throughout.
- **R13, R16 and R17 are not seeded.** Report-total mismatches, ODI copy-of-MIPS differences and statistical outliers can be created by editing rows in the simulator, but the starting day does not raise them.
- **Assumed integration shape.** Synthetic keys, Oracle Fusion field names and the 21-trays-per-stack rule are assumptions listed in docs/EVENT_CONTRACT.md section 6, to be confirmed with Bimbo IT.
- **Simulated decisions are isolated.** Planner decisions made in the simulation go to output/sim/ and never change the rule confidence learned from real data.

13. Glossary

Term	Meaning
BUSS	Customer sales-order system feeding MIPS.
OCloud Region2	Oracle Cloud supply-plan source feeding MIPS.
MIPS	Production planning system: what to make for each sales day.
ODI	Oracle Data Integrator feed of production results (finished goods, batches).
WMS	Warehouse management system: orders, allocation, tickets, loading and shipping.
Oracle Fusion	Inventory system of record; posts WMS transactions.
Seed	The synthetic starting day published into an empty event log.
Scenario	A scripted sequence of source changes that recreates one requirement scenario.
Flag ID	Rule + key, e.g. R05-900102; the engine's unique identifier for a finding.
Blind line	A schedule line with zero quantity and zero output that still sits in a live run.
Ghost run	The zero-quantity copy a moved run leaves on its original line.
Recovery run	An extra run ordered to make up a shortfall.
Load scan	The WMS scan that records goods going onto a trailer (Loaded Qty).
Derived column	A column the source system computes itself (derive.py).
Synthetic key	A row identifier the client extract lacks, added by the event contract.